Cambridge Assessment International Education

Cambridge International AS & A Level

MATHEMATICS **9709/41**

Paper 4 Mechanics **May/June 2022**

MARK SCHEME

Maximum Mark: 50

Published

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination. It shows the basis on which Examiners were instructed to award marks. It does not indicate the details of the discussions that took place at an Examiners' meeting before marking began, which would have considered the acceptability of alternative answers.

Mark schemes should be read in conjunction with the question paper and the Principal Examiner Report for Teachers.

Cambridge International will not enter into discussions about these mark schemes.

Cambridge International is publishing the mark schemes for the May/June 2022 series for most Cambridge IGCSE, Cambridge International A and AS Level and Cambridge Pre-U components, and some Cambridge O Level components.

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Generic Marking Principles

These general marking principles must be applied by all examiners when marking candidate answers. They should be applied alongside the specific content of the mark scheme or generic level descriptors for a question. Each question paper and mark scheme will also comply with these marking principles.

GENERIC MARKING PRINCIPLE 1:
Marks must be awarded in line with:

- the specific content of the mark scheme or the generic level descriptors for the question
- the specific skills defined in the mark scheme or in the generic level descriptors for the question
- the standard of response required by a candidate as exemplified by the standardisation scripts.

GENERIC MARKING PRINCIPLE 2:
Marks awarded are always **whole marks** (not half marks, or other fractions).

GENERIC MARKING PRINCIPLE 3:
Marks must be awarded **positively**:

- marks are awarded for correct/valid answers, as defined in the mark scheme. However, credit is given for valid answers which go beyond the scope of the syllabus and mark scheme, referring to your Team Leader as appropriate
- marks are awarded when candidates clearly demonstrate what they know and can do
- marks are not deducted for errors
- marks are not deducted for omissions
- answers should only be judged on the quality of spelling, punctuation and grammar when these features are specifically assessed by the question as indicated by the mark scheme. The meaning, however, should be unambiguous.

GENERIC MARKING PRINCIPLE 4:
Rules must be applied consistently, e.g. in situations where candidates have not followed instructions or in the application of generic level descriptors.

GENERIC MARKING PRINCIPLE 5:
Marks should be awarded using the full range of marks defined in the mark scheme for the question (however; the use of the full mark range may be limited according to the quality of the candidate responses seen).

GENERIC MARKING PRINCIPLE 6:
Marks awarded are based solely on the requirements as defined in the mark scheme. Marks should not be awarded with grade thresholds or grade descriptors in mind.

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Mathematics Specific Marking Principles	
1	Unless a particular method has been specified in the question, full marks may be awarded for any correct method. However, if a calculation is required then no marks will be awarded for a scale drawing.
2	Unless specified in the question, answers may be given as fractions, decimals or in standard form. Ignore superfluous zeros, provided that the degree of accuracy is not affected.
3	Allow alternative conventions for notation if used consistently throughout the paper, e.g. commas being used as decimal points.
4	Unless otherwise indicated, marks once gained cannot subsequently be lost, e.g. wrong working following a correct form of answer is ignored (isw).
5	Where a candidate has misread a number in the question and used that value consistently throughout, provided that number does not alter the difficulty or the method required, award all marks earned and deduct just 1 mark for the misread.
6	Recovery within working is allowed, e.g. a notation error in the working where the following line of working makes the candidate's intent clear.

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Mark Scheme Notes

The following notes are intended to aid interpretation of mark schemes in general, but individual mark schemes may include marks awarded for specific reasons outside the scope of these notes.

Types of mark

M Method mark, awarded for a valid method applied to the problem. Method marks are not lost for numerical errors, algebraic slips or errors in units. However, it is not usually sufficient for a candidate just to indicate an intention of using some method or just to quote a formula; the formula or idea must be applied to the specific problem in hand, e.g. by substituting the relevant quantities into the formula. Correct application of a formula without the formula being quoted obviously earns the M mark and in some cases an M mark can be implied from a correct answer.

A Accuracy mark, awarded for a correct answer or intermediate step correctly obtained. Accuracy marks cannot be given unless the associated method mark is earned (or implied).

B Mark for a correct result or statement independent of method marks.

DM or **DB** When a part of a question has two or more 'method' steps, the M marks are generally independent unless the scheme specifically says otherwise; and similarly, when there are several B marks allocated. The notation DM or DB is used to indicate that a particular M or B mark is dependent on an earlier M or B (asterisked) mark in the scheme. When two or more steps are run together by the candidate, the earlier marks are implied and full credit is given.

FT Implies that the A or B mark indicated is allowed for work correctly following on from previously incorrect results. Otherwise, A or B marks are given for correct work only.

- A or B marks are given for correct work only (not for results obtained from incorrect working) unless follow through is allowed (see abbreviation FT above).
- For a numerical answer, allow the A or B mark if the answer is correct to 3 significant figures or would be correct to 3 significant figures if rounded (1 decimal place for angles in degrees).
- The total number of marks available for each question is shown at the bottom of the Marks column.
- Wrong or missing units in an answer should not result in loss of marks unless the guidance indicates otherwise.
- Square brackets [ ] around text or numbers show extra information not needed for the mark to be awarded.

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Abbreviations

AEF/OE	Any Equivalent Form (of answer is equally acceptable) / Or Equivalent
AG	Answer Given on the question paper (so extra checking is needed to ensure that the detailed working leading to the result is valid)
CAO	Correct Answer Only (emphasising that no 'follow through' from a previous error is allowed)
CWO	Correct Working Only
ISW	Ignore Subsequent Working
SOI	Seen Or Implied
SC	Special Case (detailing the mark to be given for a specific wrong solution, or a case where some standard marking practice is to be varied in the light of a particular circumstance)
WWW	Without Wrong Working
AWRT	Answer Which Rounds To

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Question	Answer	Marks	Guidance
1(a)	$200 = \frac{(0+25)}{2}t$	**M1**	For use of $s = \frac{(u+v)}{2}t$ or other complete method to find t e.g. $v^2 = u^2 + 2as$ followed by $v = u + at$ N.B. $a = 1.5625$
	$t = 16\,\text{s}$	**A1**	
		2	
1(b)	Trapezium	**B1**	Through (0, 0) and positive t-axis
	All correct through (0, 0), (16, 25), (32, 25), (37, 0)	**B1**	FT *their* value of t from part **(a)** (t +16 + 5)
		2	
1(c)	Total distance = $200 + 400 + \frac{1}{2} \times 25 \times 5$ [= 662.5]	**M1**	Or trapezium $\frac{1}{2} \times 25(16+37)[= 662.5]$ Or $200 + 400 + \frac{25+0}{2} \times 5$ Allow *their* value of t from part **(a)** in calculating 200 and *their* time from the constant speed section from part **(b)** in calculating 400 Allow *their a* from part **(a)** if used in calculating 200
	Average speed = $\frac{662.5}{37} = 17.9\ \text{m s}^{-1}$ (3 s.f.)	**A1**	Allow $\frac{1325}{74} = 17\frac{67}{74}$
		2	

Question	Answer	Marks	Guidance
2	Use of Newton's second law for P or Q or system	**M1**	Allow sign errors. Do not allow g missing
	For P: $10 - 0.5g - T = 0.5a$, For Q: $T - 0.3g = 0.3a$, For system: $10 - 0.5g - 0.3g = 0.8a$	**A1**	For any two correct
	For attempt to solve for either a or T	**M1**	From equation(s) with no missing or extra terms. Allow g missing.
	$a = 2.5$ ms^{-2}	**A1**	
	$T = 3.75$ N	**A1**	
		5	

Question	Answer	Marks	Guidance
3	For attempt at resolving horizontally or vertically	**M1**	Allow sin/cos mix. Allow sign error. Allow g missing. Correct number of terms.
	$R = 300g - 0.28X$ or $R = 300g - X\sin 16.3$	**A1**	$\alpha = 16.26\ldots$
	$0.96X - F = 0$ or $0.96X - 0.5(300g - X\sin\alpha) = 0$ Or $X\cos 16.3 - F = 0$ or $X\cos 16.3 - 0.5(300g - X\sin\alpha) = 0$	**A1**	Or using *their* F
	Use of $F = 0.5R$	**M1**	Use to get an equation in X only. Allow sin/cos mix. Allow sign error. Allow g missing. Must be from 2 term R, which is a linear combination of $300(g)$ and a component of X
	$X = 1360$ [1363.63…]	**A1**	
		5	

Question	Answer	Marks	Guidance
4	Attempt to resolve in any direction	**M1**	For resolving. Allow sin/cos mix. Allow sign error. Correct number of terms.
	$F\cos\alpha - 20\cos 40 - 100\sin 20 = 0$ [$F\cos\alpha = 15.320\ldots + 34.202\ldots = 49.5229\ldots$]	**A1**	
	$F\sin\alpha + 20\sin 40 - 100\cos 20 = 0$ [$F\sin\alpha = 93.969\ldots - 12.855\ldots = 81.1135\ldots$]	**A1**	
	$F = \sqrt{(49.5229\ldots)^2 + (81.1135\ldots)^2}$	**M1**	OE; Attempt to solve for F; one term missing in total
	$\alpha = \tan^{-1}\left(\frac{81.1135\ldots}{49.5229\ldots}\right)$	**M1**	OE; Attempt to solve for α; one term missing in total
	$F = 95(.0)$, $\alpha = 58.6$	**A1**	$F = 95.0364\ldots$ and $\alpha = 58.5943\ldots$
	Alternative mark scheme for question 4: For candidates who use cosine and/or sine rule		
	Attempt at cosine rule from triangle of forces	**M1**	Must use lengths 100 and 20 with a suitable angle
	$F^2 = 100^2 + 20^2 - 2\times 100\times 20\cos 70$	**A1**	Correct
	$F = 95[.0]$	**A1**	
	$\frac{95.0364}{\sin 70} = \frac{20}{\sin\beta}$ OR $\frac{95.0364}{\sin 70} = \frac{100}{\sin\gamma}$	**M1**	Attempt at sin rule
		A1	where $\beta = (70 - \alpha)$ where $\gamma = (40 + \alpha)$
	$\alpha = 58.6$	**A1**	$\alpha = 58.5943\ldots$

Question	Answer	Marks	Guidance
4	**Alternative mark scheme for question 4: For candidates who resolve in other directions**		
	Attempt to resolve (e.g. parallel or perpendicular to 100 N)	**M1**	For resolving. Allow sin/cos mix. Allow sign error. Correct number of terms.
	$F\sin(\alpha+20)+20\sin 20-100=0 \quad [F\sin(\alpha+20)=93.159\ldots]$	**A1**	
	$F\cos(\alpha+20)-20\cos 20=0 \quad [F\cos(\alpha+20)=18.793\ldots]$	**A1**	
	$F=\sqrt{93.159\ldots^2+18.793^2}$	**M1**	OE; Attempt to solve for F; one term missing in total
	$\alpha=\tan^{-1}\left(\frac{93.159\ldots}{18.793\ldots}\right)-20$	**M1**	OE; Attempt to solve for α; one term missing in total
	$F = 95[.0]$, $\alpha = 58.6$	**A1**	$F = 95.0364\ldots$ and $\alpha = 58.5943\ldots$
		6	

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Question	Answer	Marks	Guidance
5(a)	For attempt at work energy equation	**M1**	3 terms. Allow sign errors. M0 for (constant) acceleration method
	$4500d - 75\,000 = \frac{1}{2} \times 1200 \times 25^2$ [= 375 000]	**A1**	Correct equation
	$d = 100$	**A1**	**AG** Accept verification with d substituted in above line to show LHS = 375 000 or LHS −RHS = 0 If no marks scored allow SCB1 for $\frac{1}{2} \times 1200 \times 25^2$
		3	
5(b)	$25^2 = 0 + 2a \times 100$ [leading to $a = 3.125$]	**B1**	Allow B1 if acceleration found in part **(a)** as 3.125 and used or stated here
	$3200 - 1200 = m \times 3.125$	**M1**	Newton's second law with 3 terms. Allow sign errors and *their a*.
	Mass of car B = 640 kg	**A1**	
	Alternative mark scheme for question 5(b)		
	For attempt at work energy equation	**M1**	3 terms. Allow sign errors.
	$(3200 - 1200) \times 100 = \frac{1}{2} \times m \times 25^2$	**A1**	Correct equation
	Mass of car B = 640 kg	**A1**	
		3	

Question	Answer	Marks	Guidance
5(c)	At P power = 3200 × 25 [= 80 000]	**B1**	For use of power = Fv
	$\frac{80000}{v} - 1200 = 0$	**M1**	Attempt Newton's second law for car B with $a = 0$ Allow *their* 80 000 (dimensionally correct)
	Steady speed = $66.7\ \text{m s}^{-1}$	**A1**	Allow $\frac{200}{3} = 66\frac{2}{3}$
		3	

Question	Answer	Marks	Guidance
6(a)	For attempt at integration	***M1**	The power of t must increase by 1 with a change of coefficient in at least 1 term. Allow if k is omitted.
	$k\left(t^3 - \frac{1}{2}t^4\right)[+C]$	**A1**	Allow unsimplified
	$k\left(8 - \frac{1}{2}\times 16\right)[-0]$	**DM1**	For use of limits 0 and 2 or substituting $t = 2$ OR equate to 0, then solve a quartic equation in t. Allow if k is omitted.
	0	**A1**	AG, CWO including stating $C = 0$ if not using limits
		4	

Question	Answer	Marks	Guidance
6(b)	$v = 0$ when $k\left(3t^2 - 2t^3\right) = 0$	**M1**	For solving for t
	Leading to $t = 1.5$ [or $t = 0$]	**A1**	A0 for other solutions not discarded
	$a = k\left(6t - 6t^2\right)$	**M1**	For differentiation, the power of t must decrease by 1 with a change of coefficient in at least 1 term. Allow if k is omitted
	$[-13.5 = k\left(6\times1.5 - 6\times1.5^2\right) \Rightarrow -13.5 = k\left(9 - 13.5\right) \Rightarrow] \; k = 3$	**A1**	
	Distance from $t = 0$ to $t = 1.5$ is $\left[3\left(t^3 - \frac{1}{2}t^4\right)\right]_0^{1.5} = 3\left(1.5^3 - \frac{1}{2}\times1.5^4\right) - 0$	**DM1**	For use of limits. $2\times\int_0^{1.5}\ldots$ or $2\times\int_{15}^{2}\ldots$ or both integrals. Dependent on first M in part **(a)**, unless they restart
	[= 2.53125] So total distance = 2 × 2.53125 = 5.06 m	**A1**	Allow distance $= \frac{81}{16} = 5\frac{1}{16}$ If DM0 then SCB1 for 5.06 without working
		6	

Question	Answer	Marks	Guidance
7(a)	$0.4\times3+0.2\times2=0.4\times2.5+0.2v$	**M1**	Use of conservation of momentum with 4 terms. Allow sign errors.
	$v = 3$ ms^{-1}	**A1**	Allow M1A0 if g included with the masses.
		2	
7(b)	For A $\pm0.4g\sin30°=0.4a$ or for B $\pm0.2g\sin30°=0.2a$ or $\pm mg\sin30°=ma$	**M1**	For either. Allow sin/cos mix.
	$a=\pm5$ or $\pm g\sin30°$	**A1**	Allow $g\sin30°$ without working for M1A1
	For B when hits barrier $v^2=3^2+2\times5\times1.6$ $[\Rightarrow v=5]$ OR $v=u+at\Rightarrow v=3+5\times0.4$ $[\Rightarrow v=5]$	**M1**	Using *their* $a\neq\pm g$ and *their* v from part **(a)** OR: use of $s=\frac{u+v}{2}t$ $1.6=\frac{3+v}{2}\times0.4[\Rightarrow v=5]$ OR $\frac{1}{2}\times0.2\times v^2-\frac{1}{2}\times0.2\times3^2=0.2\times1.6\times g\sin30$
	Speed after hitting barrier = 0.1 × 5 = 0.5	**A1**	**AG**
	v_A = 2.5 + 5 × 0.44 [= 4.7] v_B = −0.5 + 5 × 0.04 [= −0.3] or v_B = 0.5 + (–5) × 0.04 [= 0.3]	***M1**	Use of $v=u+at$ for either with correct t-value, with initial speeds ±2.5 or ±0.5 *their* $\pm a\neq\pm g$
	0.4 × 4.7 + 0.2 × (−0.3) = 0.6 v_{comb}	**DM1**	Use of $v=u+at$ for BOTH with correct t-values, initial speeds $\pm2.5,\pm0.5$ and $\pm$ *their* acceleration (same for both) and use of conservation of momentum with correct number of terms. Allow sign errors.
	v_{comb} = 3.03 ms^{-1}	**A1**	Allow $v=\frac{91}{30}=3\frac{1}{30}$ Allow DM1A0 if g included with the masses.
		7	